

“Objectifying Outcomes in Facial Aesthetic Surgery Using Artificial Intelligence”

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Short Running Head: Artificial Intelligence for Objective Aesthetic Surgery Assessment

Abstract:**Background:**

Choosing which procedure or combination of procedures is appropriate remains a daunting task for patients and surgeons alike. The goal of this study was to leverage machine learning models (MLMs) to find objective AI-derived parameters associated with improved cosmetic outcomes in patients undergoing aesthetic procedures.

Methods:

A dataset from the American Society of Plastic Surgeons (ASPS) Before and After photos was compiled for facelift, browlift, and eyelid surgeries using non-consecutive, curated photographs of all available entries. Two MLMs were trained to detect perceived age and attractiveness using “in the wild” reference images. Pre- and post-operative photographs were used to generate scores before and after surgery per patient. Uni- and multi-variate odds ratio (OR) analyses were performed to interrogate procedures and demographic factors associated with improvements in age perception, attractiveness, and a composite aesthetic benefit score.

Results:

676 patients operated on by 65 surgeons from 22 different states/provinces in the USA and Canada were included in the analyses. 328 (48.5%), 172 (25.4%), 82 (12.1%), and 427 (63.2%) underwent facelift, necklift, browlift, or eyelid surgeries, respectively. On multivariate analysis [OR (CI), p-value] older perceived preoperative age was associated with a greater combined aesthetic benefit independent of procedure type [**1.04 (1.02-1.06), p < 0.001**]. The effects of surgery were found to be additive, with patients undergoing combined procedures (facelift+necklift+browlift+eyelid) demonstrating the greatest composite aesthetic benefit [**11.34 (1.84-103.34), p < 0.05**].

Conclusions:

Machine learning can be used to more objectively assess aesthetic facial surgery using perceived age and attractiveness as novel preoperative markers. Perceived older preoperative age, decreased preoperative attractiveness, and increased number of concurrent procedures were associated improved cosmetic benefit. However, the interaction between perceived preoperative age and attractiveness demonstrated a complex, non-linear relationship when optimizing for postoperative aesthetic benefit.

Clinical Question/Level of Evidence: Level III- Retrospective Multi-Center Cohort Study

Key Words: Aging; Aesthetic Surgery; Artificial Intelligence; Machine Learning; Beauty

Introduction:

Cosmetic surgery encompassed a nearly 146-billion-dollar global market in 2024 that will continue to grow in the future.¹ The goal of facial rejuvenation surgery and related procedures is to combat the natural changes associated with aging, including loose skin, dyspigmentation, fat volume loss, and changes in facial contour.² Patients expect reliable outcomes that reduce their perceived age and improve self-confidence.³ There is also growing demand from younger patients seeking preventative treatments driven in part by social media influences and changes in public attitudes related to aesthetic surgery.⁴⁻⁶ It is the surgeon's responsibility to determine which interventions are most appropriate to meet these goals; knowing which individual or combination of procedures to recommend given the changing markets and high patient expectations remains a daunting task.

Machine learning and artificial intelligence applications have emerged as powerful tools in plastic surgery, with use cases ranging from automating image analysis for operative planning^{7,8} to skin cancer tracking⁹ and frostbite assessment.¹⁰ In cosmetic surgery, computer vision has been validated as a tool for assessing perceived change in age after periorbital rejuvenation.¹¹ Convolutional neural network models including *FaceX* have been effectively used to determine “years regained” after facial rejuvenation surgery.¹² Recently, the developers of the deep learning model *FaceAge* have used AI-generated biological age markers as tools for prognostication.¹³ Despite these promising results, there remains a lag between the evolving capacities of emergent MLMs and clinical applications.

The primary goal of this study was to explore the utility of MLMs for objectively analyzing perceived age and attractiveness before and after aesthetic facial surgeries at scale. We used publicly available outcomes from the American Society of Plastic Surgeons (ASPS) Before and

After photo gallery as a reference given its breadth of outcomes and accessibility for both surgeons and patients. We secondarily aimed to identify variables associated with decreased perceived age and improved attractiveness, including both traditional demographic and surgical factors as well as novel AI-derived scoring signals.

Methods:

Machine Learning Model Training

Two MLMs were trained to detect perceived age and perceived attractiveness. Our age classifier algorithm was trained sequentially on over 400,000 annotated reference faces using a deep multi-layer perceptron approach. Perceived attractiveness classification was trained using Likert scaling between 0-100 based on publicly available reference photos and averaged pulled scores from over 10,000 reference photos. Please refer to **Document 1, Supplemental Digital Content 1** for a comprehensive overview of model training and implementation. An overview for age perception training can be seen in **Figure 1**.

Ensemble Architecture

Two separate ensemble pipelines were constructed for perceived age and attractiveness, respectively. Age estimation was treated as a 101-class classification task using standardized resized images with augmentation for robustness. Frozen feature extractors were first constructed based on FaRL (ViT-B/16) and ResNet-50 models to construct fixed embeddings per image; the ensemble model was constructed by combining the features of these two frozen backbones. The features were projected into a common 512-dimensional space, concatenated, and fed into a 5-layer multi-layer perceptron (MLP) head. Training focused on minimizing cross entropy loss. Perceived attractiveness was trained on concatenated embeddings using three frozen backbones [FaRL (ViT-B/16), Segformer, and ResNet-50] projected into a 512-dimensional space before

being subjected to 5-layer MLP. Training was focused on minimizing MAE, with regression to a normalized continuous attractiveness score ranging from 0-1, rescaled to 0-100 for usability.

AI Model Reference Demographics and Performance

Four datasets [UTKFace, IMDB-Clean, AgeDB, and LAGENDA] were used for training our age detection model. The composite dataset contained a range of 0-116 years of age, with roughly balanced gender [worst balance = IMBD-Clean, 60/40 (%M/%F)]. Of note, race was only well-characterized in the UTKFace dataset, with a distribution of 55.9% White, 51.1% Black, 19.0% Asian, and 0.02% for Indian and Other, respectively. The state of the art (SOTA) MAE for age ranged between 3.65 to 5.55 years, with our own MAE achieving 3.07 years.

For our perceived attractiveness model, six models (Face Research Lab London, American Multiracial Faces, Chicago Face, Hawaii Face, MEBeauty, MR2, and SCUT- FBP5500) with mixed ordinal rating scales (1-7 or 1-10) were rescaled to construct a composite dataset with an ordinal scale of 0-100. Four of the six datasets had near perfect gender balance of 50/50 (+/-) 1%. The reported racial demographic distribution varied significantly. All included images had a minimum of 15 and maximum of 60 raters per image. The SOTA MAE ranged from 0.2 (SCUT-FBP5500, 1-7 rating) to 0.55 (American Multiracial Faces, 1-7 rating); our model's MAE achieved 6.8 (1-100 rating) as seen in **Table 1, Supplemental Digital Content 2**.

Data Collection

A retrospective study was performed using curated, non-consecutive photos obtained from all available ASPS Before and After photographs for patients undergoing facelift, necklift, browlift, and eyelid surgeries. Of note, this dataset only contains curated outcomes from submitting surgeons. Only patients with complete descriptions related to their primary procedure were included. Additional information such as time to postoperative follow-up, age, gender, operating

surgeon, and geography was identified. All patients were sub-categorized based on their primary procedure and/or combination procedures. Primary procedures were defined as facelift, necklift, browlift, or eyelid surgery, including canthopexies, lower, and upper blepharoplasties.

AI- Generated Parameters

Pre- and postoperative photographs were used to generate individual age and attractiveness scores before and after surgical intervention. Our primary outcomes of interest were change in perceived attractiveness scores and change in perceived age before and after surgical intervention. A composite normalized z-score (aesthetic benefit score) was also constructed accounting for changes in both perceived attractiveness and age. When the actual pre-operative age was available, a biological age signal was calculated using the difference in perceived and chronological age divided by chronological age.

Fairness Evaluation by Relevant Subgroups

Given the potential for bias inherent when evaluating attractiveness, sub-analyses for sensitivity and model performance of perceived attractiveness were carried out for stratified age, gender, and race. Performance parameters included Pearson's r and MAE (Confidence Intervals). Testing parameters per group included Kruskal-Wallis H testing and Cohen's d evaluation. Pairwise Wilcoxon Rank Sum testing was applied for pairwise comparisons.

Procedure and Patient Characteristics Associated with Changes in Perceived Attractiveness and Age

The changes in perceived age, attractiveness, and aesthetic benefit scores were calculated for each patient undergoing individual or combined procedures. Shapiro-Wilk testing was used to confirm normality of preoperative perceived age and attractiveness scoring. Subsequent analyses of the differences of perceived age, attractiveness, and combined aesthetic scoring revealed non-

normal distributions. Nonparametric Kruskal-Wallis Rank Sum testing and Pairwise Wilcoxon Rank Sum testing were used for evaluating group- and pairwise level differences in perceived age, attractiveness, and combined aesthetic scores by procedure type.

Odds Ratio Analyses and Multivariate Modeling

Univariate odds ratio analysis was performed to identify procedure- and patient-level factors associated with improvements in age, attractiveness, and aesthetic benefit scores. Relevant variables identified were subsequently included in multi-variate odds ratio modeling. Outcomes are presented as [OR, CI, p-value]. Contour plots were constructed to demonstrate the complex multivariate relationship between preoperative perceived age, attractiveness, and combined aesthetic benefit score by procedure type.

Statistical Thresholds and Figure Generation

A minimum alpha level of 0.05 was used as a threshold of significance for statistical testing unless otherwise specified. All figure generation and statistical interrogation were performed using the software R (v. 4.3.2).

Results:

Patient Characteristics

Operations performed on a total of 676 patients by 65 surgeons from 22 different states/provinces in the USA and Canada were included in the final cohort. The median (IQR) follow-up time was 3 (2-5.25) mo. Of the patients identified, 328 (48.5%), 172 (25.4%), 82 (12.1%), and 427 (63.2%) underwent facelift, necklift, browlift, or eyelid surgeries, respectively, with the majority of patients undergoing a combination of procedures. The majority of those with reported genders were female 290 (90.3%). While not explicitly stated, race was inferred in the dataset using the *FairFace* model developed by Karkkainen *et. al.*¹⁴ The majority of patients in

the dataset were white 496 (73.4%), followed by Asian (14.1%), and Latino/Hispanic (8.1%). An overview of relevant demographic and patient-specific characteristics can be seen in **Table 1**.

Fairness Evaluation of Attractiveness by Subgroups

On subgroup analyses, there were statistically significant differences in perceived attractiveness by gender [Kruskal-Wallis $H= 5.69$, ($p<0.01$)], although the differences were not associated with clinically significant changes in model performance based on Cohen's d testing ($d=0.18$).

Notably, there were no significant differences in model performance when comparing outcomes by inferred race or preoperative perceived age. A comprehensive evaluation of model performance by subgroup is included in **Table 2, Supplemental Digital Content 3**.

Changes in Perceived Age and Perceived Attractiveness by Procedure

Of the patients undergoing individual procedures, the greatest average change (median [IQR]) in perceived age was associated with facelift (-4.46 [-7.65 - (-0.31)] yrs) compared with eyelid (-2.12 [-7.5 - 1.12] yrs) or browlift (-0.39 [-3.75 - 1.81] yrs) surgery alone. The greatest associated changes were seen with combined surgeries including browlift+eyelid+facelift+necklift surgery (-8.92 [-10.65 - (-2.70)] yrs), although there were no statistically significant differences on pairwise testing between groups.

When examining changes in perceived attractiveness scores after single procedures, patients undergoing facelift (2.93 [0.98 - 4.68]) surgery had the greatest relative improvement, with statistically significant differences compared to eyelid surgery (0.75 [-1.00 - 2.62]) alone ($p < 0.001$). The greatest benefit of combined procedures was appreciated after facelift+eyelid+browlift surgery (3.87 [1.90 - 4.14]), with statistically significant differences when compared to the eyelid surgery subgroup ($p < 0.001$) (**Table 3, Supplemental Digital Content 4**).

Univariate Associations of Perceived Changes of Age and Attractiveness After Surgery

On univariate OR analysis, older perceived preoperative age was proportionately associated with perceived age reduction [**1.05 (1.04-1.07), $p < 0.001$**], increased attractiveness [**1.03 (1.02-1.05), $p < 0.001$**], and combined aesthetic benefit [**1.05 (1.04-1.07), $p < 0.001$**]. Patients that looked older than 70 years old (y.o.) before surgery were 9.36 times more likely to look younger after surgery than those who looked 40 y.o. or younger (**$p < 0.001$**) before surgery. They were also 3.85 (**$p < 0.001$**) times more like to have an improved attractiveness score and 11.24 (**$p < 0.0001$**) times more likely to have an improved combined aesthetic benefit score postoperatively. When scaling for the patient's actual age, patients that looked 10 to 25% younger than their actual age were 6.8 times more likely to look younger (**$p < 0.01$**) after surgery and 8.98 times more likely to have improved aesthetic benefit (**$p < 0.01$**) than patients that looked 25% older or more than their actual age. In contrast, increased preoperative attractiveness was inversely associated with improved postoperative attractiveness [**0.93 (0.90-0.97), $p < 0.001$**] and combined aesthetic benefit [**0.94 (0.91-0.98), $p < 0.001$**]. Patients with scores higher than 30 were only 36% (**$p < 0.01$**) as likely to see a benefit in perceived attractiveness and 59% (**$p < 0.05$**) as likely to see a combined aesthetic benefit compared to those with scores between 10 to 20. Men were less likely to have an improved aesthetic result [**0.25 (0.084-0.63), $p < 0.01$**]. Follow-up time and actual preoperative age were not found to be associated with response to surgery (**Table 2**).

Multivariate Evaluation of Response to Surgery Accounting for Procedure Type

On multivariate analysis, older perceived preoperative age was associated with greater perceived age reduction [**1.06 (1.03- 1.08), $p < 0.001$**] and overall improved aesthetic benefit [**1.04 (1.02 1.06), $p < 0.001$**] controlling for procedure type and preoperative perceived attractiveness.

Patients that looked older than 70 y.o. were 10.62 (**p < 0.001**) times more likely to look younger and 6.12 (**p < 0.001**) times more likely to have an improved combined aesthetic benefit after surgery than those that looked 40 y.o. or younger. In contrast, greater perceived preoperative attractiveness was associated with lower odds of improvement in relative postoperative attractiveness [**0.90 (0.86-0.94), p < 0.001**] and combined aesthetic benefit [**0.96 (0.92-0.99), p < 0.001**]. Patients with scores higher than 30 were only 28% (**p < 0.01**) and 91% (**p < 0.05**) as likely to see improvement in perceived attractiveness or combined aesthetic benefit, respectively. Scaling perceived age with actual age, patients that looked more than 25% younger than their actual age were 17.4 times more likely to have a combined aesthetic benefit from surgery compared to patients that looked 25% older or more than their actual age (**p < 0.01**) independent of preoperative perceived age, attractiveness, or procedure.

Compared to browlift surgery, patients undergoing eyelid procedures alone were more likely to look younger [**2.72 (1.003-7.42) p < 0.05**]. After facelift surgery, patients were more likely to look younger [**3.23 (1.13-9.35), p < 0.05**], appear more attractive [**8.26 (2.82-25.25), p < 0.01**], and have a combined aesthetic benefit score [**6.15 (2.02- 23.15), p < 0.01**] than patients undergoing browlift alone. Combination procedures were independently associated with improved age perception, attractiveness, and combined aesthetic benefit. Facelift+eyelid surgery [**5.7 (1.45-26.3), p < 0.05**] was associated with the greatest odds of perceived age reduction, facelift+necklift+eyelid surgery [**13.83 (3.89-54.37), p < 0.001**] with the most improved perceived attractiveness, and facelift+necklift+browlift +eyelid surgery [**11.34 (1.84-103.34), p < 0.05**] with the most likely improved combined aesthetic benefit when compared to browlift alone, independent of other factors specified in **Table 4, Supplemental Digital Content 5**.

Maximizing Combined Aesthetic Benefit by Predicted Preoperative Age and Attractiveness

Contour plots were constructed to isolate the relative combined benefit density achieved across two dimensions, preoperative predicted age and attractiveness, for isolated facelift, eyelid, and browlift procedures. Areas highlighted in red represent areas of preoperative predicted attractiveness and age scores with the greatest relative benefit for each respective surgery. For facelift, younger appearing patients with lower perceived preoperative attractiveness scores and older appearing patients with higher attractiveness scores achieved the greatest combined aesthetic benefit after surgery (**Figure 2a**). Comparatively, older patients with lower preoperative attractiveness scores benefited the most from eyelid procedures (**Figure 2b**), while those with higher preoperative attractiveness scores benefited more from browlift surgery (**Figure 2c**). A three-dimensional representation of the facelift contour plot was constructed in **Figure 3a**, with the global minimum and maximum highlighted. The vignette in **Figure 3b-c** demonstrates an optimal facelift outcome; the patient's perceived age decreased from 54.4 to 44.1 y.o. while the attractiveness score increased from 41.7 to 51.7 comparing pre- and postoperative frontal views photos.

Discussion:

AI and machine learning applications continue to expand to more specialized use cases in medicine. In aesthetic surgery, computer vision can serve as a third-party proxy for objective perceived age evaluation.^{11,12,15} As the demand for facial surgery in younger patients seeking subtle, preventative effects expands, perceived attractiveness may also prove to be an important AI signal.

Previous investigators have used robust datasets with racially diverse samples, large rater pools, and a standardized Likert scoring systems to create predictive MLMs for assessing beauty or

attractiveness objectively.^{16–18} Likewise, we sought to construct robust models for perceived age and attractiveness capable of generating complimentary signals for objective outcome evaluation in the broadening aesthetic demographic.

We recognize the sensitivity of constructing standardized scores for attractiveness and age considering the complex relationship between perceived attractiveness and social determinants including race.^{19,20} Given the heterogenous nature of the data used to train our models, we used *FairFace*¹⁴ for racial inference. Subsequent model interrogation revealed no performances biases comparing prediction scores by race. Previous work by authors *Chatterjee et al.* have explored the complex relationship between aging, perceived attraction, and inherent biases of the rater, with older perceived faces tending to elicit strong value judgements.²¹ We encourage readers to approach our results with caution, as the ASPS data included is not necessarily balanced or representative of the greater aesthetic population. It is important to note that the models developed herein were designed to inform surgical decision-making and patient counseling and not value judgement.

When comparing outcomes between individual procedures, facelift surgery had the greatest relative benefit in perceived age and attractiveness compared to browlift or eyelid surgery alone. The combination of multiple procedures also proved to have a synergistic effect on decreased perceived age, increased perceived attractiveness, and increased overall aesthetic benefit, a trend that held true on multivariate analysis when controlling for other relevant factors. It is unclear why browlift alone had relatively less impactful outcomes, although factors including operative technique merit further investigation.

Of note, there was a positive association between older preoperative perceived age and greater perceived age reduction after surgery. This finding parallels previous studies in the cosmetic

space including the relative benefit of hyaluronic acid fillers²² as well as the greater patient satisfaction of older patients after cosmetic facial surgery.²³ In contrast, timing for optimal surgical intervention remains divisive. Previous studies of patients undergoing facelift surgeries have demonstrated superior cosmetic results in younger patients.²⁴ Another study demonstrated later time to revision after facelift surgery in patients younger than 53.²⁵ Such findings were often based on limited sample sizes, pointing to a need for more objective evaluations in larger patient populations with longer follow-ups.

In our study, we found that higher perceived preoperative attractiveness scores were associated with reduced relative improvements in attractiveness or combined aesthetic benefit independent of perceived preoperative age. When examining the effects of perceived preoperative age and attractiveness concurrently on combined aesthetic benefit, a more complex relationship arises; optimization varied significantly by procedure type as demonstrated by the contour plots. While increased perceived preoperative age was generally associated with better outcomes, patients that appeared significantly older than their actual age tended to have worse outcomes than their younger appearing counterparts. These findings parallel emerging evidence that biological rather than chronological age is a more important predictor of health outcomes in other clinical contexts.²⁶ Evidence of premature aging identified by our MLM may reflect underlying genetic, health, or environmental factors that predispose certain patients to poorer aesthetic surgical outcomes. More importantly, the strong associations detected by our model were identified independent of operator oversight; these signals could serve as an important flag for patients not likely to benefit from surgery that would be missed otherwise. For example, patients with excessive prior sun exposure causing skin texture changes and dyschromia may not

see the same relative benefit in our model without adjunctive chemical or laser-based treatments, although this topic requires further investigation.

While these preliminary findings are informative, additional prospective analysis in a more controlled setting is warranted for validating our AI-driven signals before more widespread implementation is considered.

Limitations:

The authors recognize several key limitations. The accuracy of the annotations is dependent on the available public data included. We assume that variables including patient age, gender, and procedure type were accurately provided to the ASPS by the submitting surgeon; there was no feasible way of cross-referencing these data. Missing data including preoperative patient age and gender could not be accounted for. Additional interventions including skin resurfacing through laser and chemical treatments often performed concurrently and may affect relative improvement appreciated postoperatively. As the photos are specifically uploaded to represent surgical outcomes, we are limited in our ability to comment on these effects specifically. Future work can focus on the relative aesthetic benefit of these interventions in isolation from or conjunction with operative interventions.

It is also important to note that the authors had no control over which specific case examples that were uploaded. We should conservatively assume that the outcomes represent a subsample of the “best” expected outcomes as an important selection bias with a likely inflationary effect on perceived aesthetic and age improvement, constraining the generalizability of the paper.

Our median time to follow-up was limited to three months postoperatively. For most cosmetic procedures, it can take up to year for full resolution of postoperative edema and facial

normalization.²⁷ Future studies could investigate the natural evolution of perceived attractiveness and age over time, especially as it related to different surgical approaches.

The training of the models included in this study were informed by well-established approaches highlighted in **Document 1, Supplemental Digital Content 1**. Training sources included robust datasets with highly variable lighting, camera quality, and viewing angles; see **Document 2, Supplemental Digital Content 6** for a summary reference to all models and techniques referred to and implemented in this manuscript. While the majority of uploaded pre and postoperative photos follow pre-established professional photography angles, the variability of picture quality could influence outcome evaluations. Future investigations could benefit from using plastic surgery raters as standard references.

While evaluation of attractiveness is subjective, there is burgeoning interest in creating more standardized scales to objectify this parameter in plastic surgery. Regardless of technique, there is always going to be inherent bias associated with how attractiveness is either defined or perceived. For younger patients seeking preventative benefit, longitudinal postoperative photos are necessary before commenting on the relative robustness of any given technique.

Conclusions:

Machine learning and artificial intelligence applications can be used to objectify previously subjective outcome measurements. Moreover, they can help us identify novel parameters, including the difference in perceived and actual age, previously inaccessible associated with significant effects on aesthetic outcomes.

Compliance with Ethical Standards

Institutional Review Board Approval:

The study protocol received approval from our Institutional Review Board (IRB) under the reference number 23-006815.

Statement of Human Rights and Informed Consent:

Informed consent was waived due to the nature of the study. Images were sourced from the publicly accessible ASPS gallery where patient consent was obtained for public display; informed consent for the representative case vignette was collected per institutional policy. All reference data for model training and validation is available through open-source publicly available repositories.

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Figures

Figure 1. Machine Learning Training Overview for Perceived Age

A composite pre-annotated dataset composed of over 390,000 faces pre-annotated for age was compiled for training and validation (top, left). The reference photos separated into training (80%) and validation (20%) subsets (top, middle). Data augmentation was performed using a series of techniques to improve robustness and generalizability, including random ordering/removal, rotation, flipping, etc (middle, left).

Ensemble modeling training was used for performance optimization using weights cross-referenced from FaRL and ResNet-50 models (left, bottom). A composite deep multi-layer perceptron model was constructed, optimized, and validated using 20% of the allocated dataset (right, bottom).

Figure 2. Contour Plots for Combined Aesthetic Benefit by Perceived Age and Preoperative Attractiveness

Areas represented in red represent population subsets achieving greater improvements in combined aesthetic benefit score, while areas in blue represent the contrary. The x and y axis correlate to preoperative perceived preoperative age and attractiveness. This was constructed for patients undergoing isolated:

- a) Facelift Surgery
- b) Eyelid Surgery (lower or upper blepharoplasty, canthopexy)
- c) Browlift Surgery

Figure 3. Three-Dimensional Plot of Combined Aesthetic Score Optimization for Facelift Surgery

- a) A representative reprojected three-dimensional contour plot demonstrates local maximums and minimums of combined aesthetic score, with the maximum and minimum highlighted.
- b) Oblique views from case vignette with perceived preoperative (52.9 y.o., 39.8) and postoperative (41.7 y.o., 45.1) age, attractiveness scores on the left and right respectively.
- c) Frontal views from case vignette with perceived preoperative (54.4 y.o., 41.7) and postoperative (44.1 y.o., 51.7) age, attractiveness scores on the left and right respectively.

Tables

Table 1. Summary Statistics and Demographics

Summary Statistics are provided for available demographic and surgical distributions.

Table 2. Univariate OR Assessment of Surgical Benefit

Univariate OR analyses were performed for improved perceived age and attractiveness as well as and combined aesthetic benefit. Terms in **bold** represent statistically significant effects.

Supplemental Digital Material Legend

Document 1, Supplemental Digital Content 1

This document provides a high-level overview of the relevant techniques employed for model training both age and attractiveness estimations.

Table 1, Supplemental Digital Content 2

This table summarizes relevant demographic and performance level statistics for the training data repository references for Perceived Age [UTKFace, IMBD-Clean, AgeDB, and LAGENDA] and Perceived Attractiveness [Face Research Lab London, American Multiracial Faces, Chicago Face, Hawaii Face, MEBeauty, MR2, and SCUT-FBP5500]. Only UTKFace had available racial

demographics.¹² Relevant gender, age, and race demographics are included based on availability. State of the art performance metrics are provided for Perceived Age (years) and Perceived Attractiveness (ordinal rating) as mean average errors. References to the manuscripts for the datasets are provided.

Table 2, Supplemental Digital Content 3

This table demonstrates summary results from sub-analyses of performance and sensitivity by gender, *FairFace*-inferred race, and gender. for Perceived Age [UTKFace, IMBD-Clean, AgeDB, and LAGENDA] and Perceived Attractiveness [Face Research Lab London, American Multiracial Faces, Chicago Face, Hawaii Face, MEBeauty, MR2, and SCUT-FBP5500]. Outputs include per variable Pearson's r and mean average errors with interquartile ranges. Kruskal-Wallis H testing was calculated for comparing between group, with Cohen's d representing realized difference by group when applicable. Pairwise Wilcoxon Rank Sum Testing was included when applicable when comparing multiple groups simultaneously.

Table 3, Supplemental Digital Content 4

Results are provided for change in perceived age, attractiveness, and combined aesthetic benefit by procedure type and for combination of procedures. Kruskal-Wallis Rank Sum testing was performed to determine if differences between groups was detected per outcome parameter. Pairwise Wilcoxon Rank Sum Testing was subsequently performed per group to evaluated for pairwise differences detected between procedure types. Terms in **bold** represent statistically significant effects.

Table 4, Supplemental Digital Content 5

Multivariate OR modeling was performed controlled for perceived preoperative age, attractiveness, procedure type, and the normalized difference in perceived age and actual age.

Terms in **bold** represent statistically significant effects.

Document 2, Supplemental Digital Content 6

This document provides a summary legend of all supplemental digital content provided for review.

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Table 1. Summary Statistics and Demographics

Characteristic:	Descriptive Statistic
Surgeons: (n)	65
States/Provinces: (n)	22
Patients: (n)	676
Gender: (n, %)	
Female	290 (90.3)
Male	31 (9.3)
Undefined	355
Race (n, %) *	
White	496 (73.4)
Asian	95 (14.1)
Latino/Hispanic	55 (8.1)
Black	16 (2.4)
Middle Eastern	10 (1.5)
Indian	4 (0.6)
Age: (yr, n=335, Mean, SD)	56.75 (11.3)
Follow-Up Time: (mo, Median, n=232, IQR)	3.00 (2.00-5.25)
Procedures: (n, %)	
Facelift	328 (48.5)
Necklift	172 (25.4)
Browlift	82 (12.1)
Eyelid	427 (63.2)

Table 2. Univariate OR Assessment of Surgical Benefit

Characteristic:	Younger Appearance		Benefit Improved Attractiveness Score		Combined Benefit	
	OR (95% CI)	p-value	OR (95% CI)	p-value	OR (95% CI)	p-value
Preoperative Age						
Younger	REF	-	REF	-	REF	-
Older	1.00 (0.98-1.03)	0.47	1.008 (0.99-1.03)	0.41	1.01 (0.99-1.03)	0.46
Perceived Preoperative Age						
-	REF	-	REF	-	REF	-
Change in Odds Per Year Increase	1.05 (1.04-1.07)	< 0.001	1.03 (1.02-1.05)	< 0.001	1.05 (1.04-1.07)	< 0.001
Perceived Preoperative Age, Groups (y.o.)						
Age ≤ 40	REF	-	REF	-	REF	-
40 > Age ≤ 50	2.50 (1.33-4.76)	< 0.01	2.07 (1.10-3.92)	< 0.05	4.49 (2.01-11.49)	< 0.001
50 > Age ≤ 60	3.19 (1.74-5.93)	< 0.001	2.01 (1.10-3.68)	< 0.05	5.46 (2.50-13.72)	< 0.001
60 > Age ≤ 70	5.29 (2.81-10.12)	< 0.001	3.45 (1.85-6.46)	< 0.001	10.20 (4.66-25.71)	< 0.0001
Age > 70	9.36 (3.91-24.64)	< 0.001	3.85 (1.77-8.70)	< 0.001	11.24 (4.63-30.76)	< 0.0001
Δ(Actual Age-Perceived Age)/Actual Age						
n ≥ 0.25	REF	-	REF	-	REF	-
0.1 ≥ n < 0.25	2.44 (0.79-8.02)	0.12	1.76 (0.56-5.53)	0.32	3.53 (0.882-23.75)	0.11
-0.1 < n < 0.1	3.33 (1.14-10.39)	< 0.05	2.47 (0.85-7.38)	0.095	5.16 (1.37-33.63)	< 0.05
-0.25 > n < -0.1	6.83 (1.99-2.55)	< 0.01	2.42 (0.74-8.10)	0.14	8.98 (2.18-61.42)	< 0.01
n ≤ -0.25	6.38 (0.018-1.75)	0.98	2.9 (0.78-11.71)	0.11	16.71 (3.50-126.60)	< 0.01
Perceived Preoperative Attractiveness						
-	REF	-	REF	-	REF	-
Change in Odds Per Point Increase	0.99 (0.95-1.03)	0.74	0.93 (0.90-0.97)	< 0.001	0.94 (0.91-0.98)	< 0.001
Perceived Preoperative Attractiveness, Groups						
10 > Score ≤ 20	REF	-	REF	-	REF	-
20 > Score ≤ 30	1.01 (0.61-1.64)	0.95	0.42 (0.23-0.73)	< 0.01	0.88 (0.56-0.83)	< 0.05
Score > 30	0.87 (0.46-1.66)	0.69	0.36 (0.17-0.71)	< 0.01	0.59 (0.32-0.68)	< 0.05
Gender						
Female	REF	-	REF	-	REF	-
Male	2.46 (0.23-1.06)	0.067	0.59 (0.28-1.24)		0.25 (0.084-0.63)	< 0.01
Follow-up Time (mo)						
Shorter	REF	-	REF	-	REF	-
Longer	0.96 (0.92-1.001)	0.059	0.96 (0.92-1.005)	0.0984	0.99 (0.95-1.03)	0.57

Figure 1

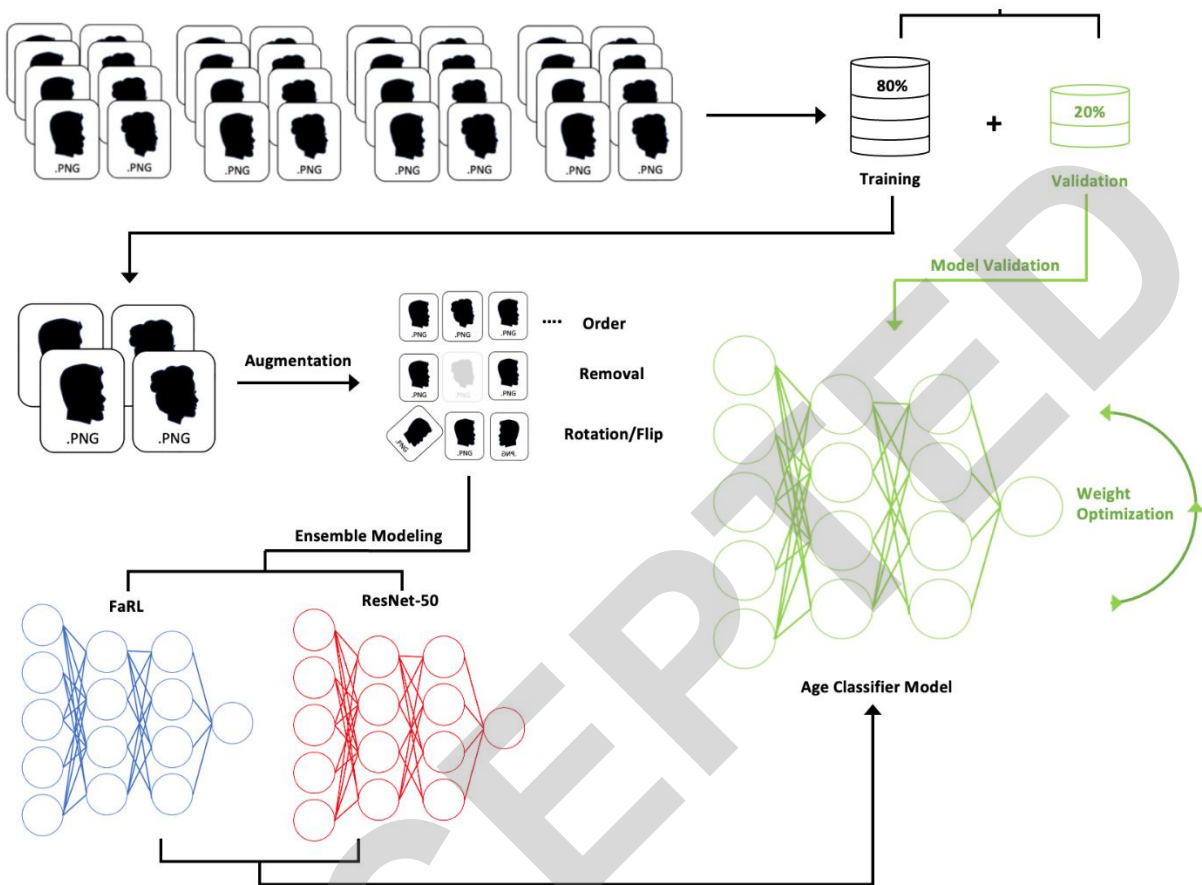


Figure 2a

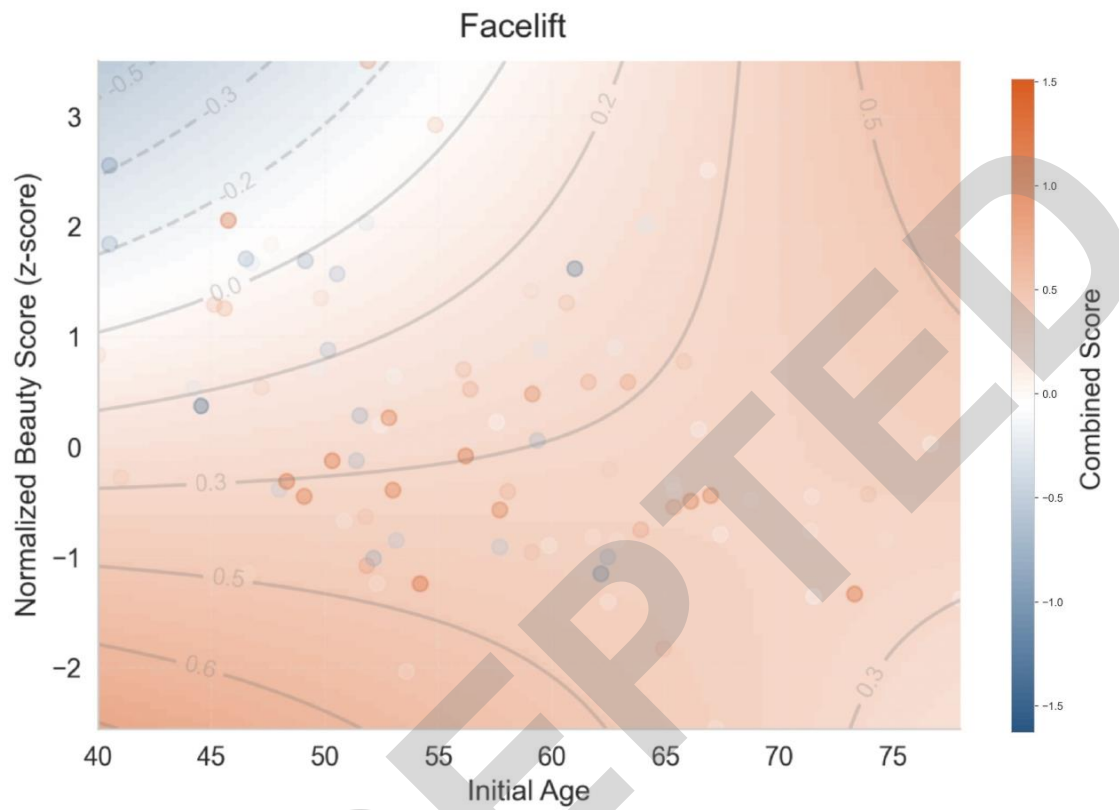


Figure 2b

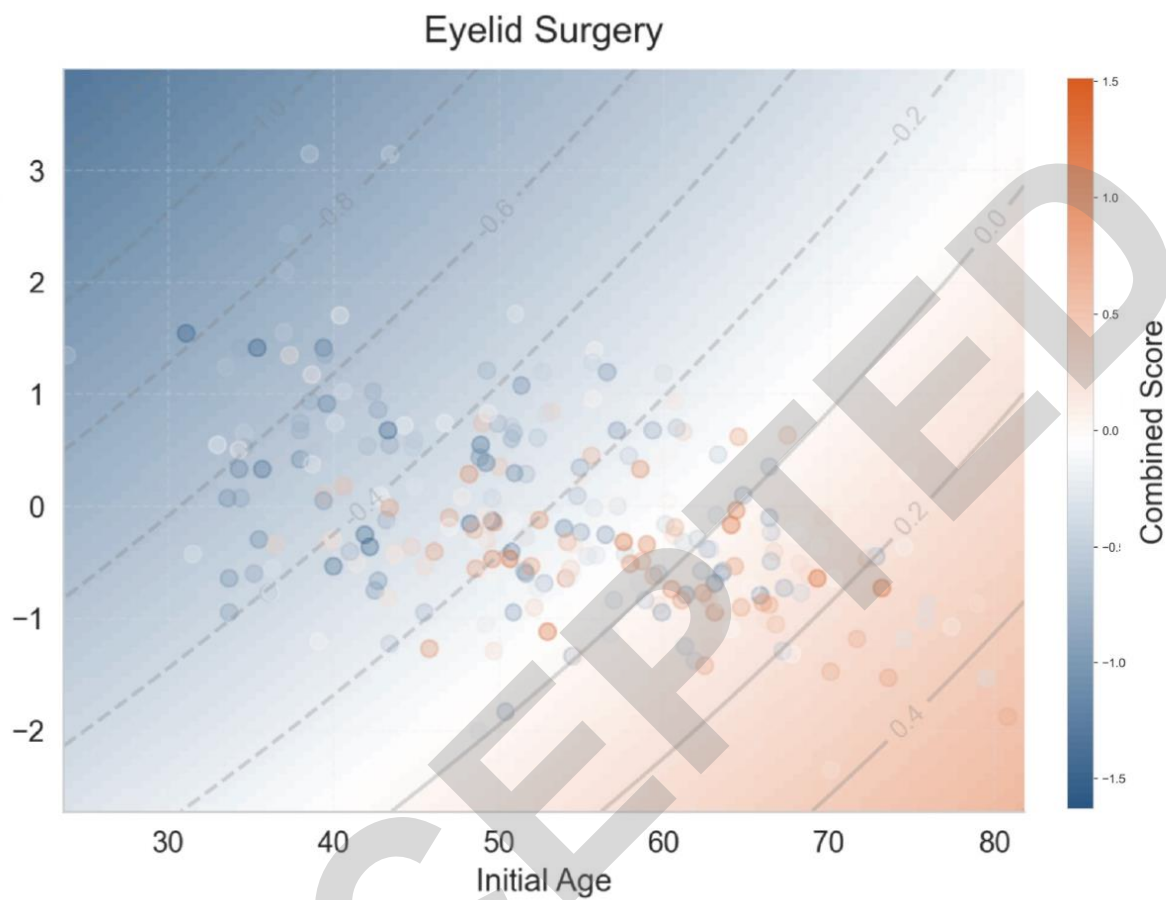


Figure 2c

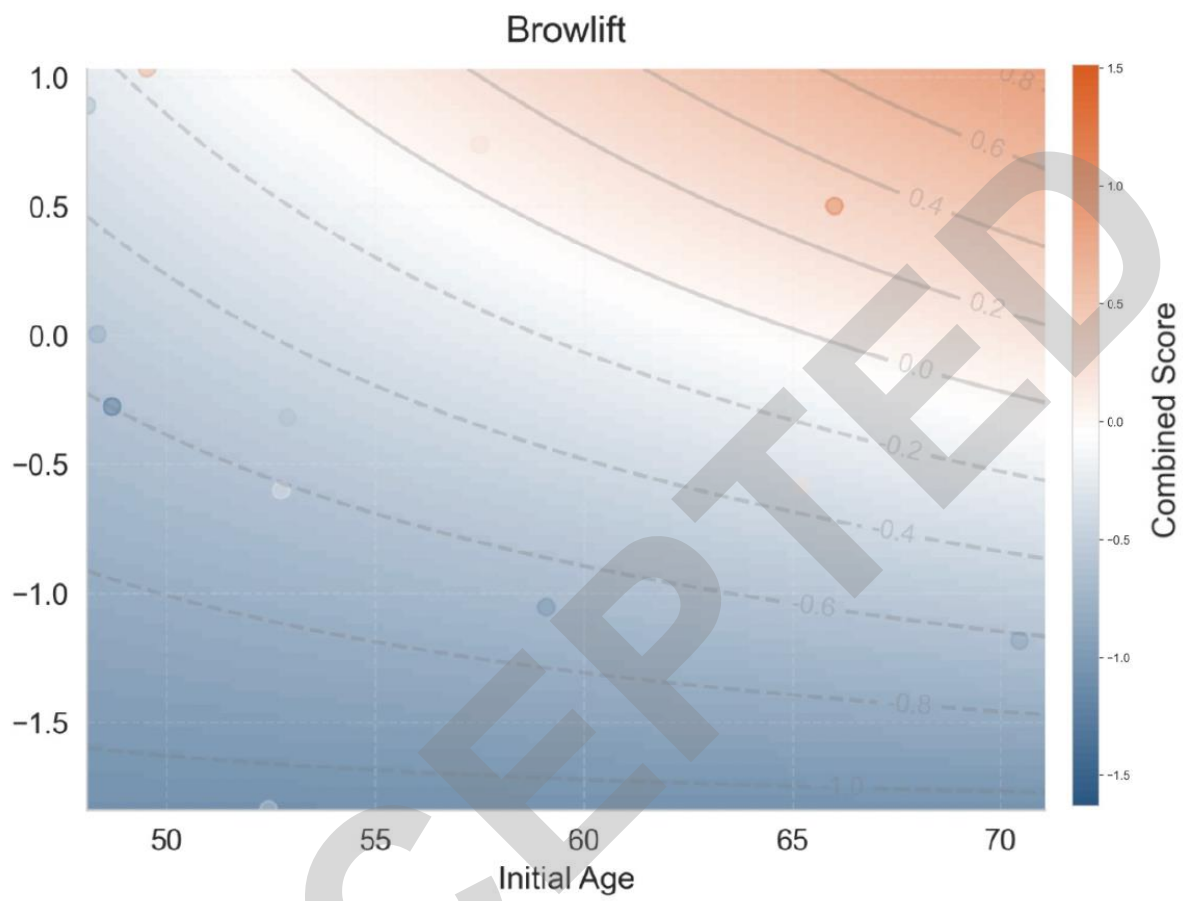


Figure 3a

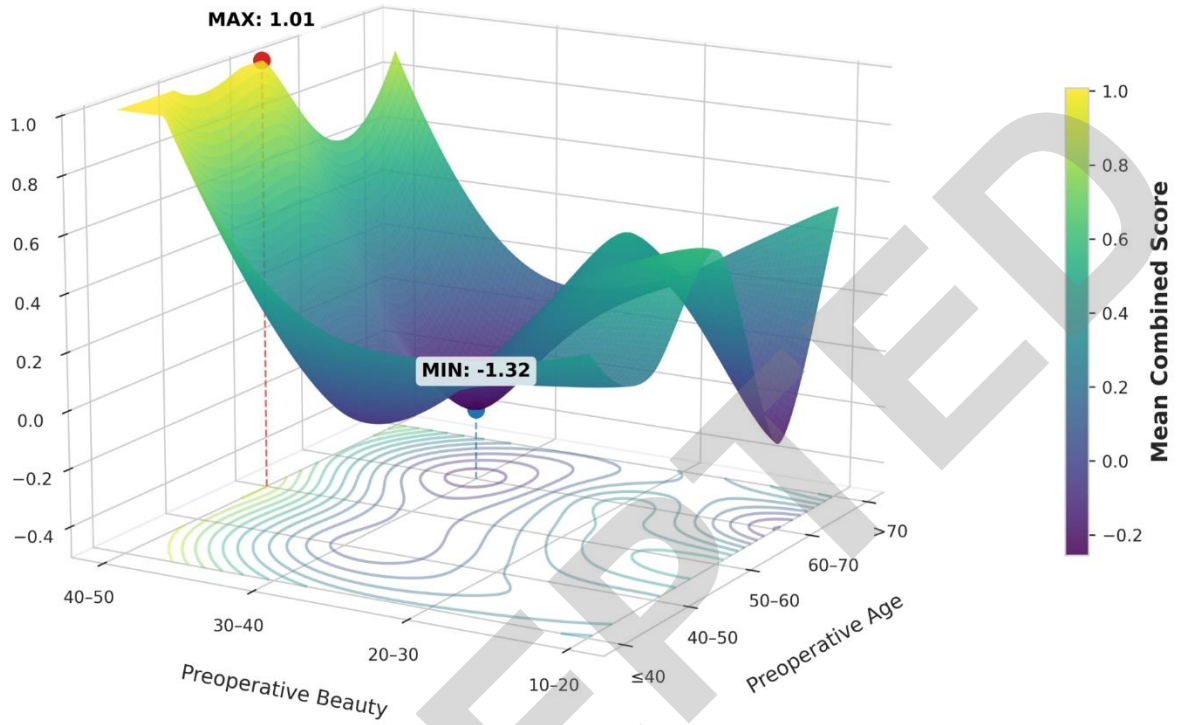


Figure 3b



Figure 3c



SDC 1. Supplemental Material 1: Model Training and Validation

Overview

We trained two models:

- (1) Perceived Age
- (2) Perceived Attractiveness.

Below we summarize pipelines, datasets, and evaluation.

Common Pipeline

Feature extractors (frozen):

Large, publicly available backbones produce fixed image embeddings. Model-specific backbones are listed below.

Data curation & splits:

Faces are standardized (e.g., 224×224) and augmented. We use fixed training/validation/testing splits with a reproducible seed to avoid leakage. Normalizers and statistics are fit on training data only.

Head & objective:

A lightweight MLP is trained on the frozen features. Optimization uses first-order methods with early stopping based on held-out validation performance.

Perceived Age

Task:

Age estimation framed as 101-class classification over ages 0–100; evaluated with mean absolute error (MAE, years).

Architecture:

An ensemble combines two frozen backbones: FaRL (ViT-B/16; 512-d) and ResNet-50 (2048-d) pretrained on face-attribute data including FairFace. Following prior work emphasizing large-scale pretraining over loss tweaks, we prioritized robust pretrained representations. Each backbone's features are projected to 512-d, concatenated (1024-d), and passed to a 5-layer MLP with GELU, BatchNorm, spatial and progressive dropout to produce 101 logits.

Training & Evaluation:

The head is trained with CrossEntropyLoss using AdamW, weight decay, and a schedule with linear warmup then reduce-on-plateau. The best validation checkpoint is selected. We track cross-entropy and MAE; the final model attains MAE = 3.07 years on the held-out test set.

Augmentation: Random horizontal flips, RandAugment, and random erasing.

Training data: UTKFace (>20k faces, ages 0–116); IMDB-Clean (cleaned IMDB-WIKI);

AgeDB (16,488 images, ages 1–101); LAGENDA (67,159 images, ages 0–95).

Perceived Attractiveness

Task:

Regression to a continuous score normalized to [0,1].

Architecture:

A trainable MLP head over concatenated embeddings from three frozen backbones: FaRL (512-d), SegFormer-B5 face-parsing features (49-d), and ResNet-50 (2048-d). Each is projected to 512-d; concatenated features feed a 5-layer MLP with GELU, LayerNorm, and progressive dropout.

Training & Evaluation:

Minimize L1 (MAE) with AdamW; linear warmup followed by reduce-on-plateau; gradient clipping and early stopping on validation loss. Report MAE on the held-out test set.

Augmentation:

Random flips, RandAugment, random erasing.

Training data (≥ 50 raters/photo):

FRL (1,020 images), AMFD (219 images), CFD (827 identities), Hawai'i Face Database (700 images), MEBeauty (2,550 faces), MR2 (74 identities), SCUT-FBP5500 (5,500 faces).

Label Harmonization:

Datasets with different scales are linearly rescaled to [0,1] using their known endpoints; image-level mean ratings are used.

Training Results:

Early stopping (patience 15) halted at epoch 76; final training loss 0.0855, validation loss 0.0664; the model closely approximates average human ratings in MAE.

SDC 2. Supplemental Table 1. Training Data Demographics and Performance Metrics

Perceived Age: Demographic and Performance Breakdown.								
		UTKFace ¹	IMDB-Clean ²	AgeDB ³	LAGENDA ⁴			
Gender, N (%)								
	Male	12,391 (52.3)	330,652 (60)	9,471 (57.4)	42,096 (50)			
	Female	11,314 (47.7)	224,356 (40)	7,017 (42.6)	42,096 (50)			
Age, Median (Range)								
		30 (0–116)	37 (0-100)	48 (1-101)	32 (0-95)			
Race, N (%)								
	White	10,078 (55.9)	-	-	-			
	Black	4,526 (25.1)	-	-	-			
	Asian	3,434 (19.0)	-	-	-			
	Indian	3 (0.02)	-	-	-			
	Other	5 (0.02)	-	-	-			
MAE, Years (SOTA)								
		3.65	3.82	4.01	5.55			
Perceived Attractiveness: Demographic and Performance Breakdown.								
	Face Research Lab London ⁵	American Multiracial Faces ⁶	Chicago Face ⁷	Hawaii Face ⁸	MEBeauty ⁹	MR2 ¹⁰	SCUT-FBP5500 ¹¹	
Gender, N (%)								
	Male	34 (33.3)	20 (18.2)	299 (50.1)	70 (50)	1380 (60)	37 (50)	2750 (50)
	Female	68 (66.7)	90 (81.8)	298 (49.9)	70 (50)	920 (40)	37 (50)	2750 (50)
Age, Median (Range)								
	N/A (17–90)	31.6 (18–85)	29 (17–65)	21 (18–44)	25 (10-70)	27 (N/A)	N/A (15-60)	
Race, N (%)								
	White	70 (68)	-	197 (31)	-	2300 (100)	22 (30)	1500 (27)
	Black	13 (13)	-	183 (33)	-	-	32 (43)	-
	Asian	19 (19)	-	109 (18)	-	-	20 (27)	4000 (73)
	Indian	-	-	-	-	-	-	-
	Latino	-	-	108 (18)	-	-	-	-
	Other	-	-	-	-	-	-	-
	Mixed	-	110 (100)	-	-	-	-	-
	Number of Raters per Image	15 minimum	25.5 mean	30-40	20-30	60	15-20	60
MAE, Ordinal Rating (SOTA)								
	1-7	1-7	1-7	1-7	1-10	1-7	1-5	
	0.35	0.55	0.40	0.45	0.5	0.50	0.20	

SDC 3. Supplemental Table 2. Subgroup Analyses of Potential Perceived Attractiveness Biases

Model Performance and Sensitivity Analysis by Gender.			
		Pearson's r	MAE (95% CI)
Gender			
	Male	0.805	7.4 (8.0-9.2)
	Female	0.783	8.6 (6.9-8.0)
		Kruskal-Wallis H, (p)	5.69 (p=0.017)
		Cohen's d	0.18
Model Performance and Sensitivity Analysis by Race.			
Race			
	White	0.814	7.9 (7.3-8.6)
	Asian	0.827	7.6 (7.0-8.2)
	Latino/Hispanic	0.766	9.5 (7.4-11.7)
	Black	0.648	7.8 (6.5-9.1)
	Middle Eastern	0.783	8.3 (6.2-10.5)
	Indian	0.604	9.9 (8.0-12.1)
		Kruskal-Wallis H, (p)	H=7.01, (p=0.22)
		Pairwise Wilcoxon Rank Sum Testing	No significant differences between groups
Model Performance and Sensitivity Analysis by Age.			
	Age ≤ 40	0.961	5.7 (5.6-5.8)
	40 > Age ≤ 50	0.758	6.6 (6.5-6.7)
	50 > Age ≤ 60	0.684	7.4 (7.3-7.6)
	60 > Age ≤ 70	0.689	7.9 (7.7-8.0)
	Age > 70	0.784	8.5 (8.4-8.7)
		Kruskal-Wallis H, (p)	6.207 (p=0.3411)
		Pairwise Wilcoxon Rank Sum Testing	No significant differences between groups

SDC 4. Supplemental Table 3. Assessment of Perceived Change in Age, Attractiveness, and Combined Aesthetic Benefit by Procedure

Characteristic:					Age Δ, yrs: Median, [IQR]	Attractiveness Δ, Score: Median, [IQR]	Composite Δ, Z Score: Median, [IQR]
Procedure(s):							
Group 1: (n=18)	Browlift				-0.39 [-3.75-1.81]	0.00 [-1.08-1.90]	-0.59 [-0.95-(0.20)]
Group 2: (n=296)		Eyelid			-2.12 (-7.5-1.12)	0.75 [-1.00-2.62]	-0.23 [-0.73- 0.25]
Group 3: (n=114)			Facelift		-4.46 (7.65+0.31)]	2.93 [0.98-4.68]	0.15 [-0.28-0.57]
Group 4: (n=33)	Browlift	Eyelid			-3.70 [-8.30-0.60]	-0.54 [-1.70-2.56]	-0.33 [-0.61-(-0.01)]
Group 5: (n=4)	Browlift		Facelift		-0.40 [-1.90-1.72]	2.17 [1.65-3.43]	-0.12 [-0.25-(-0.03)]
Group 6: (n=29)		Eyelid	Facelift		-3.60 [-8.40-(-1.48)]	3.70 [2.42-7.00]	0.31 [-0.11-0.75]
Group 7: (n=106)			Facelift	Necklift	-2.50 [-8.03-(-0.09)]	2.75 [0.33-5.19]	0.17 [-0.47-0.55]
Group 8: (n=10)	Browlift	Eyelid	Facelift		-6.17 [-12.10-(-3.97)]	3.87 [1.90-4.14]	0.33 [-0.02-0.69]
Group 9: (n=7)	Browlift		Facelift	Necklift	-8.02 [-9.42-(-4.46)]	2.06 [-0.09-5.76]	-0.14 [-0.17-0.90]
Group 10: (n=49)		Eyelid	Facelift	Necklift	-5.43 [-7.87-(-0.66)]	3.16 [1.16-5.80]	0.26 [-0.22-0.777]
Group 11: (n=9)	Browlift	Eyelid	Facelift	Necklift	-8.92 [-10.65-(-2.70)]	3.46 [2.40-4.04]	0.52 [0.14-0.65]
Group Comparisons:					Kruskal-Wallis Rank Sum Test: (p-value)	Kruskal-Wallis Rank Sum Test: (p-value)	Pairwise Wilcoxon Rank Sum Test: (p-value)
					0.054	<0.0001	<0.0001
Pairwise Comparisons:					Pairwise Wilcoxon Rank Sum Test: (p- value)	Pairwise Wilcoxon Rank Sum Test: (p- value)	Pairwise Wilcoxon Rank Sum Test: (p- value)
Pairs:							
	Group 1		Group 10		-	-	<0.05
	Group 1		Group 6		-	-	<0.05
	Group 2		Group 3		-	<0.001	<0.001
	Group 2		Group 4		-	-	<0.01
	Group 2		Group 5		-	<0.001	-
	Group 2		Group 6		-	-	-
	Group 2		Group 7		-	-	-
	Group 2		Group 8		-	<0.001	-
	Group 2		Group 9		-	-	<0.001
	Group 2		Group 10		-	<0.001	<0.001
	Group 3		Group 4		-	<0.01	-
	Group 4		Group 7		-	-	-
	Group 4		Group 5		-	<0.01	-
	Group 4		Group 6		-	<0.01	-
	Group 4		Group 10		-	<0.01	-

SDC 5. Supplemental 4. Multivariate OR Assessment of Surgical Benefit by Preoperative Perceived Age, Attractiveness, and Procedure

Characteristic:	Younger Appearance		Benefit Improved Attractiveness		Combined Benefit	
	OR (95% CI)	p-value	OR (95% CI)	p-value	OR (95% CI)	p-value
Perceived Preoperative Age						
-	REF	-	REF	-	REF	-
Change in Odds Per Year Increase	1.06 (1.03-1.08)	< 0.001	6.53 (0.98-50.24)	0.70	1.04 (1.02-1.06)	< 0.001
Perceived Preoperative Age, Groups (y.o.)						
Age ≤ 40	REF	-	REF	-	REF	-
40 > Age ≤ 50	2.51 (1.30-4.90)	< 0.01	1.39 (0.71-2.74)	0.32	3.57 (1.56-9.28)	< 0.01
50 > Age ≤ 60	3.26 (1.69-6.36)	< 0.01	1.11 (0.58-2.15)	0.74	3.87 (1.71-9.96)	< 0.01
60 > Age ≤ 70	5.67 (2.79-11.77)	< 0.001	1.38 (0.68-2.79)	0.36	6.30 (2.73-16.47)	< 0.0001
Age > 70	10.62 (4.05-30.37)	< 0.001	1.27 (0.51-3.18)	0.60	6.12 (2.32-17.79)	< 0.0001
Perceived Preoperative Attractiveness						
-	REF	-	REF	-	REF	-
Change in Odds Per Point Increase	1.04 (0.99-1.08)	0.062	0.90 (0.86-0.94)	< 0.001	0.96 (0.92-0.99)	< 0.05
Perceived Preoperative Attractiveness, Groups						
10 > Score ≤ 20	REF	-	REF	-	REF	-
20 > Score ≤ 30	1.54 (0.88-2.64)	0.11	0.46 (0.24-0.83)	< 0.05	1.26 (0.76-2.07)	0.35
Score > 30	1.7 (0.81-3.56)	0.15	0.28 (0.12-0.61)	< 0.01	0.91 (0.45-1.89)	< 0.05
Δ(Actual Age-Perceived Age)/ Actual Age						
n ≥ (0.25)	REF	-	REF	-	REF	-
(0.1) ≥ n < (0.25)	1.81 (0.54-6.45)	0.25	1.14 (0.33-4.07)	0.83	2.44 (0.53-17.98)	0.29
(-0.1) < n < (0.1)	1.60 (0.55-5.64)	0.33	1.64 (0.47-5.79)	0.43	2.97 (0.66-21.90)	0.20
(-0.25) > n < (-0.1)	3.06 (0.77-13.11)	0.44	1.71 (0.43-6.99)	0.44	5.03 (1.01-39.49)	< 0.01
n ≤ (-0.25)	4.16 (0.68-3.04e134)	0.11	3.39 (0.78-15.50)	0.10	17.4 (3.17-148.71)	< 0.01
Procedure(s)						
Browlift	REF	-	REF	-	REF	-
Eyelid	2.72 (1.003-7.42)	< 0.05	2.11 (0.74-5.80)	0.13	2.6 (0.89-9.48)	0.10
Facelift	3.23 (1.13-9.35)	< 0.05	8.26 (2.82-25.25)	< 0.001	6.15 (2.02-23.15)	< 0.01
Facelift Necklift	3.23 (1.11-9.45)	< 0.05	5.69 (1.97-16.99)	< 0.001	5.48 (1.78-20.70)	< 0.01
Facelift Browlift	0.71 (0.071-7.09)	0.76	2.29 e7 (4.21e-49-N/A)	0.98	0.91 (0.040-10.36)	0.95
Facelift Eyelid	5.7 (1.45-26.3)	< 0.05	7.09 (1.90-29.85)	< 0.001	9.78 (2.60-44.26)	< 0.05
Browlift Eyelid	2.34 (0.70-8.06)	0.17	1.03 (0.32-3.42)	0.96	1.2 (0.31-5.24)	0.80
Facelift Necklift Browlift	3.39 (3.1e-15-N/A)	0.98	3.64 (0.56-31.96)	0.19	2.04 (0.29-14.41)	0.46
Facelift Browlift Eyelid	4.19 (0.76-33.79)	0.12	5.07 (0.92-41.17)	0.082	8.57 (1.58-58.92)	< 0.05

Facelift	Necklift	Eyelid		3.48 (1.05 - 2.01)	< 0.05	13.83 (3.89- 54.37)	< 0.001	5.66 (1.69- 22.87)	< 0.01
Facelift	Necklift	Browlift	Eyelid	5.53 (0.74- 116.08)	0.15	2.52e7 (7.40e-20- N/A)	0.98	11.34 (1.84- 103.34)	< 0.05

ACCEPTED

SDC 6. Supplemental Material 6: Legend and Supplemental Digital Material References

Supplemental Digital Material Legend

Document 1, Supplemental Digital Content 1

This document provides a high-level overview of the relevant techniques employed for model training both age and attractiveness estimations.

Table 1, Supplemental Digital Content 2

This table summarizes relevant demographic and performance level statistics for the training data repository references for Perceived Age [UTKFace, IMBD-Clean, AgeDB, and LAGENDA] and Perceived Attractiveness [Face Research Lab London, American Multiracial Faces, Chicago Face, Hawaii Face, MEBeauty, MR2, and SCUT-FBP5500]. Only UTKFace had available racial demographics.¹² Relevant gender, age, and race demographics are included based on availability. State of the art performance metrics are provided for Perceived Age (years) and Perceived Attractiveness (ordinal rating) as mean average errors. References to the manuscripts for the datasets are provided.

Table 2, Supplemental Digital Content 3

This table demonstrates summary results from sub-analyses of performance and sensitivity by gender, *FairFace*-inferred race, and gender. for Perceived Age [UTKFace, IMBD-Clean, AgeDB, and LAGENDA] and Perceived Attractiveness [Face Research Lab London, American Multiracial Faces, Chicago Face, Hawaii Face, MEBeauty, MR2, and SCUT-FBP5500]. Outputs include per variable Pearson's r and mean average errors with interquartile ranges. Kruskal-Wallis H testing was calculated for comparing between group, with Cohen's d representing

realized difference by group when applicable. Pairwise Wilcoxon Rank Sum Testing was included when applicable when comparing multiple groups simultaneously.

Table 3, Supplemental Digital Content 4

Results are provided for change in perceived age, attractiveness, and combined aesthetic benefit by procedure type and for combination of procedures. Kruskal-Wallis Rank Sum testing was performed to determine if differences between groups was detected per outcome parameter. Pairwise Wilcoxon Rank Sum Testing was subsequently performed per group to evaluated for pairwise differences detected between procedure types. Terms in **bold** represent statistically significant effects.

Table 4, Supplemental Digital Content 5

Multivariate OR modeling was performed controlled for perceived preoperative age, attractiveness, procedure type, and the normalized difference in perceived age and actual age. Terms in **bold** represent statistically significant effects.

Ethics & Licensing:

All datasets are public research sets used under their licenses; we did not attempt re-identification. Pretraining that includes FairFace and the use of diverse beauty datasets aim to mitigate demographic bias.

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